Exp. 10 Creating CI/CD Pipeline in Azure DevOps

AIM

To create and configure a Continuous Integration and Continuous Deployment (CI/CD) pipeline in Azure DevOps for automatic build and deployment of an application.

TOOLS REQUIRED

- Microsoft Azure DevOps
- GitHub or Azure Repos
- Web Browser
- Azure Subscription

PREREQUISITES

- Project source code available in GitHub
- Azure Web App already created
- Azure DevOps account and project created

PROCEDURE

Step 1: Create Azure DevOps Project

- Login to Azure DevOps portal
- Click on **New Project**
- Enter:
    - Project Name (e.g., QuizAppPipeline)
    - Visibility: Private/Public
- Click **Create**

Step 2: Connect Repository

- Go to **Pipelines → Create Pipeline**
- Select source:
  - GitHub (or Azure Repos)
- Authorize access to GitHub
- Select your repository:
  - *Online-Quiz-Management-System*

Step 3: Configure Pipeline

- Choose pipeline configuration:
  - YAML file (recommended)
- Azure will auto-detect environment (Node.js / Python)

Example YAML:

```yaml
trigger:
- main

pool:
  vmImage: 'ubuntu-latest'

steps:
- task: UseNode@1
  inputs:
    version: '18.x'

- script: |
    npm install
    npm run build
  displayName: 'Install and Build'

- task: AzureWebApp@1
  inputs:
    azureSubscription: 'Your-Azure-Connection'
    appName: 'quizapp-online'
    package: '.'
```

Step 4: Run Pipeline

- Click **Run Pipeline**
- Pipeline will:
    - Fetch code
    - Install dependencies
    - Build application
    - Deploy to Azure

Step 5: Monitor Pipeline

- View pipeline execution in **Pipelines → Runs**
- Check:
    - Build logs
    - Deployment status
- Fix errors if any

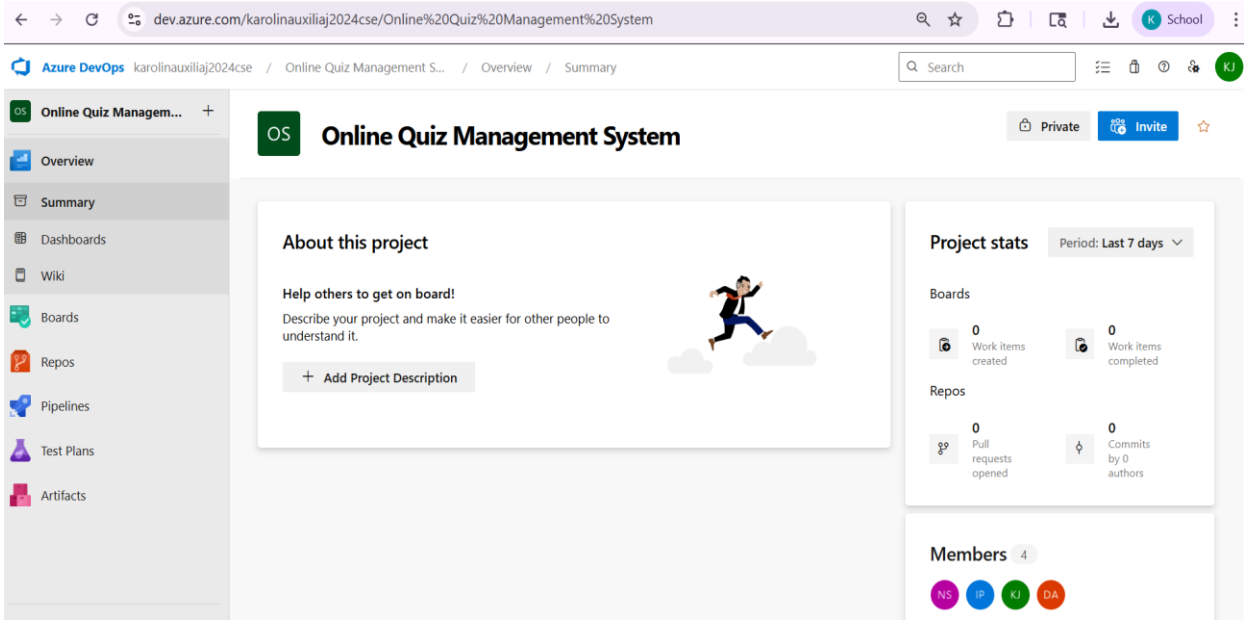
Step 6: Enable Continuous Integration

- Every time code is pushed to GitHub:
    - Pipeline triggers automatically
    - Application updates in Azure

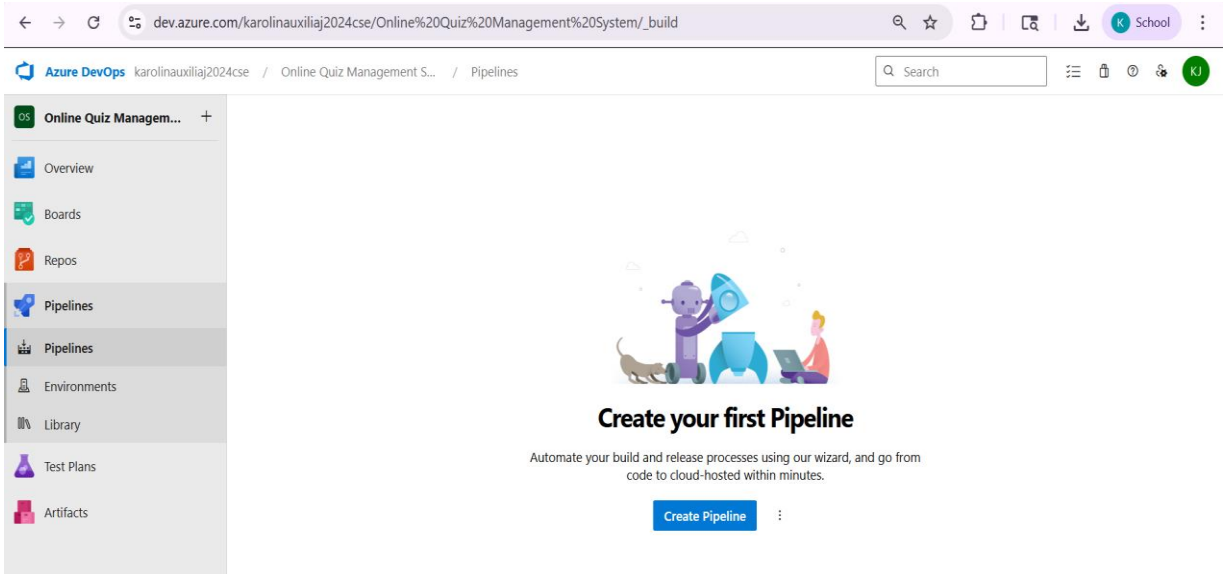
RESULT

The CI/CD pipeline was successfully created in Azure DevOps. The application was automatically built and deployed whenever changes were pushed to the repository.

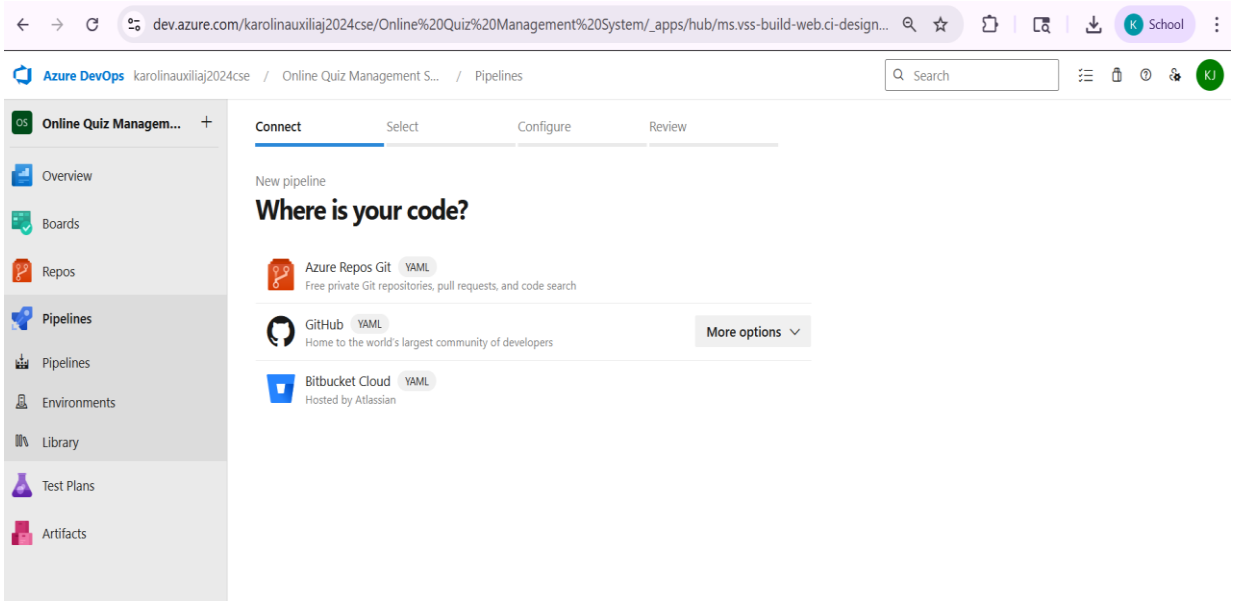
Online Quiz Management System in Azure



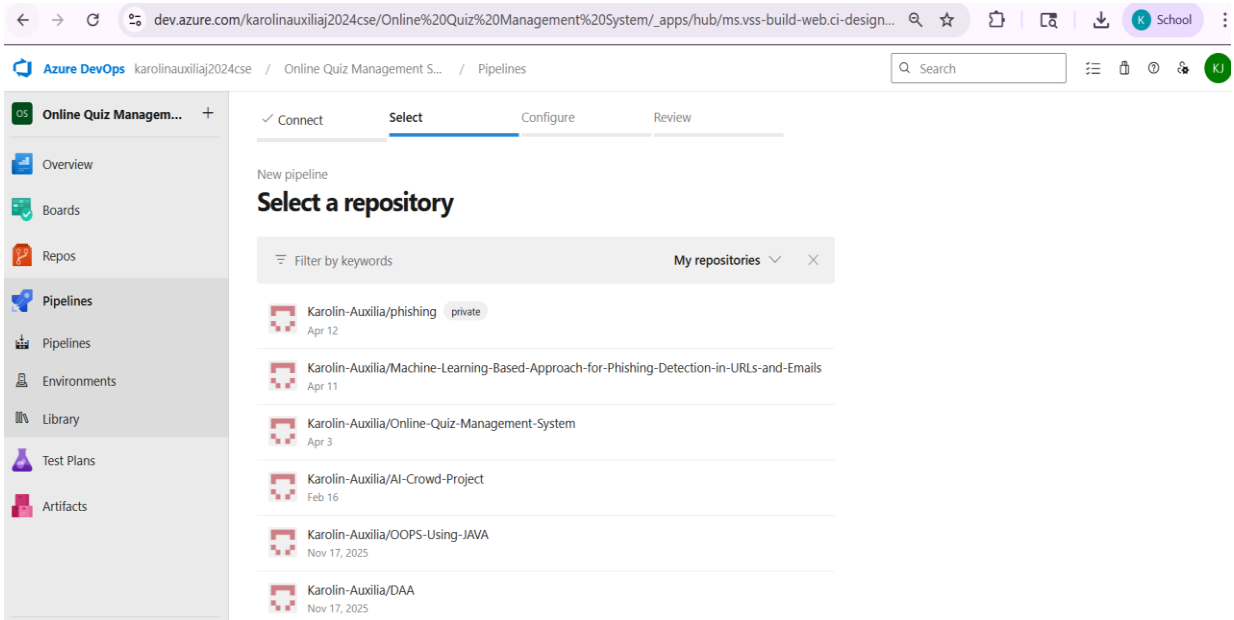
Select Pipeline



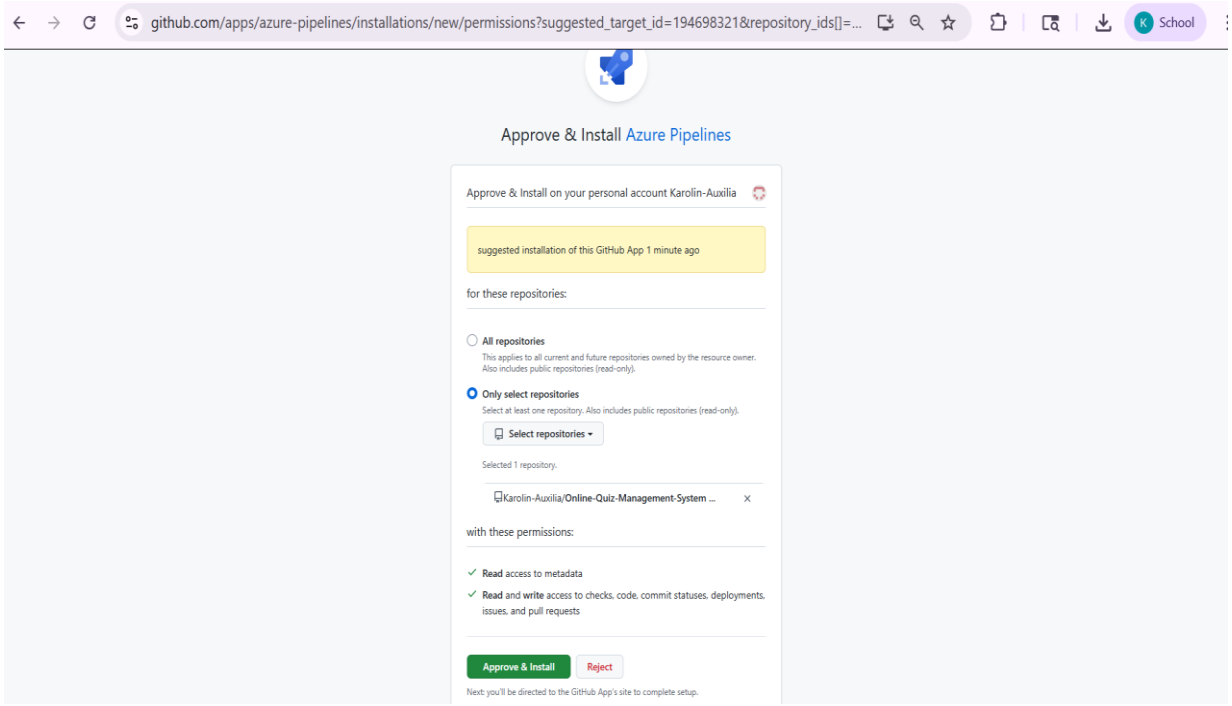
After Clicking Create Pipeline

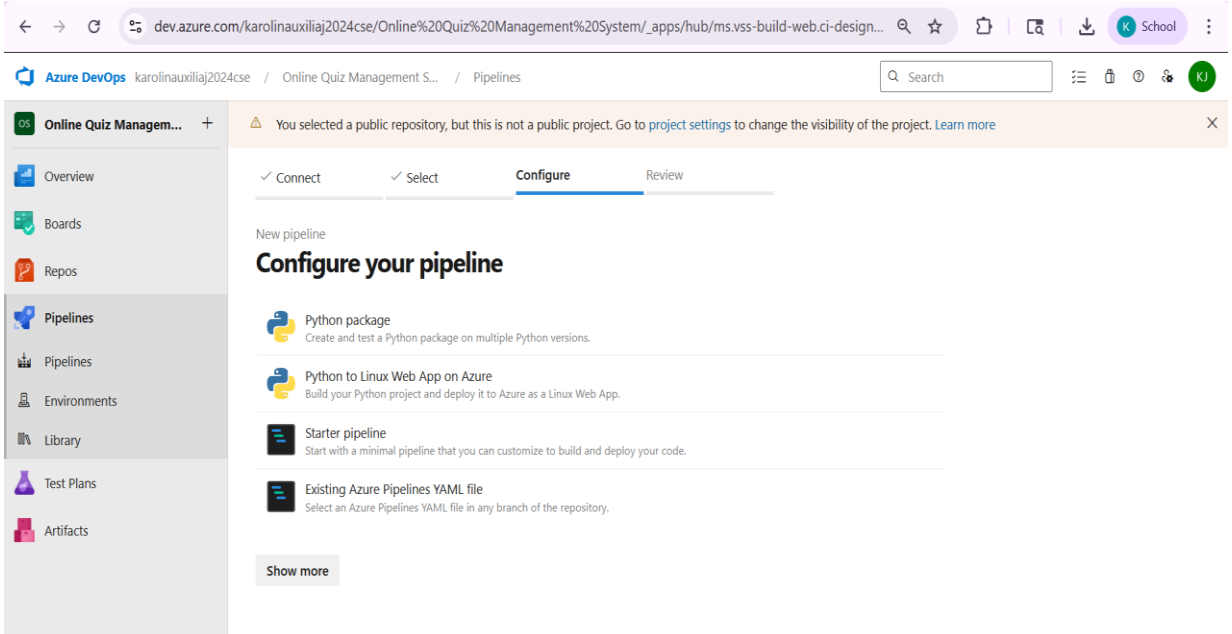


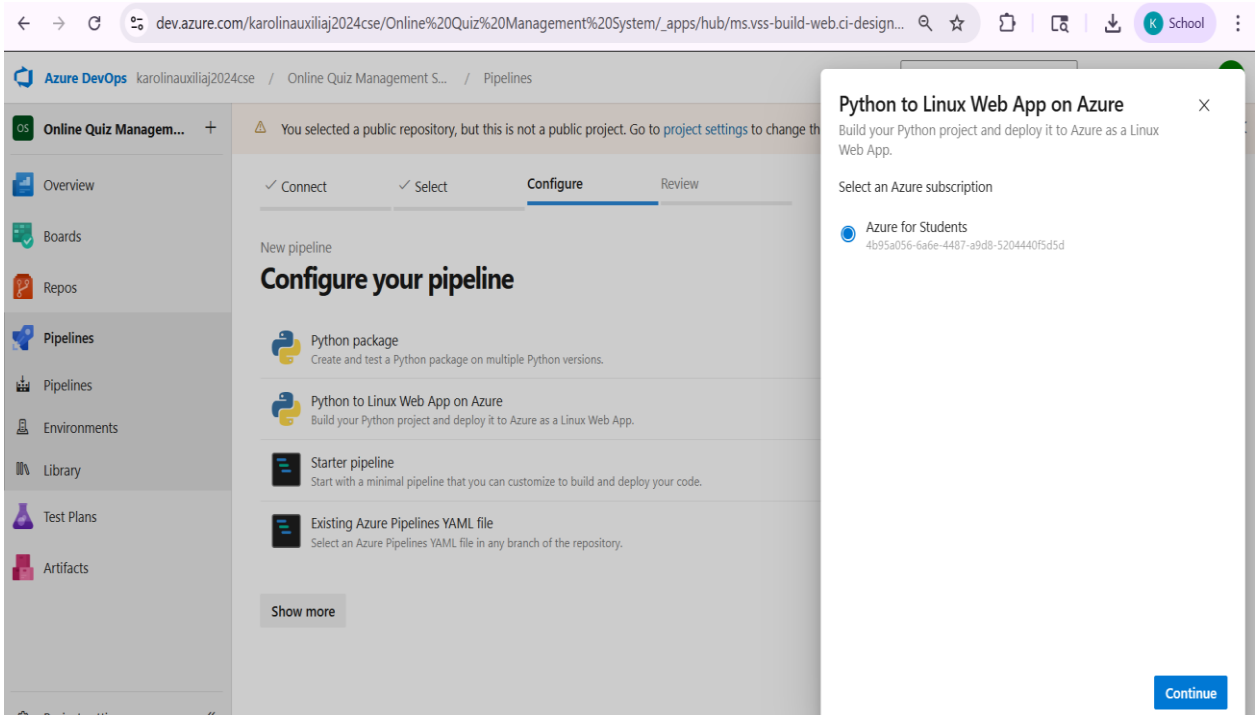
1. Select GitHub
2. After choosing GitHub
3. Select a Repository appears



Choose Online Quiz Management Repository



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You selected a public repository, but this is not a public project. Go to project settings to change th
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Select
Configure
Review
New pipeline
Configure your pipeline
Python package
Create and test a Python package on multiple Python versions.
Python to Linux Web App on Azure
Build your Python project and deploy it to Azure as a Linux Web App.
Starter pipeline
Start with a minimal pipeline that you can customize to build and deploy your code.
Existing Azure Pipelines YAML file
Select an Azure Pipelines YAML file in any branch of the repository.
Show more
Python to Linux Web App on Azure
Build your Python project and deploy it to Azure as a Linux Web App.
Select an Azure subscription
Azure for Students
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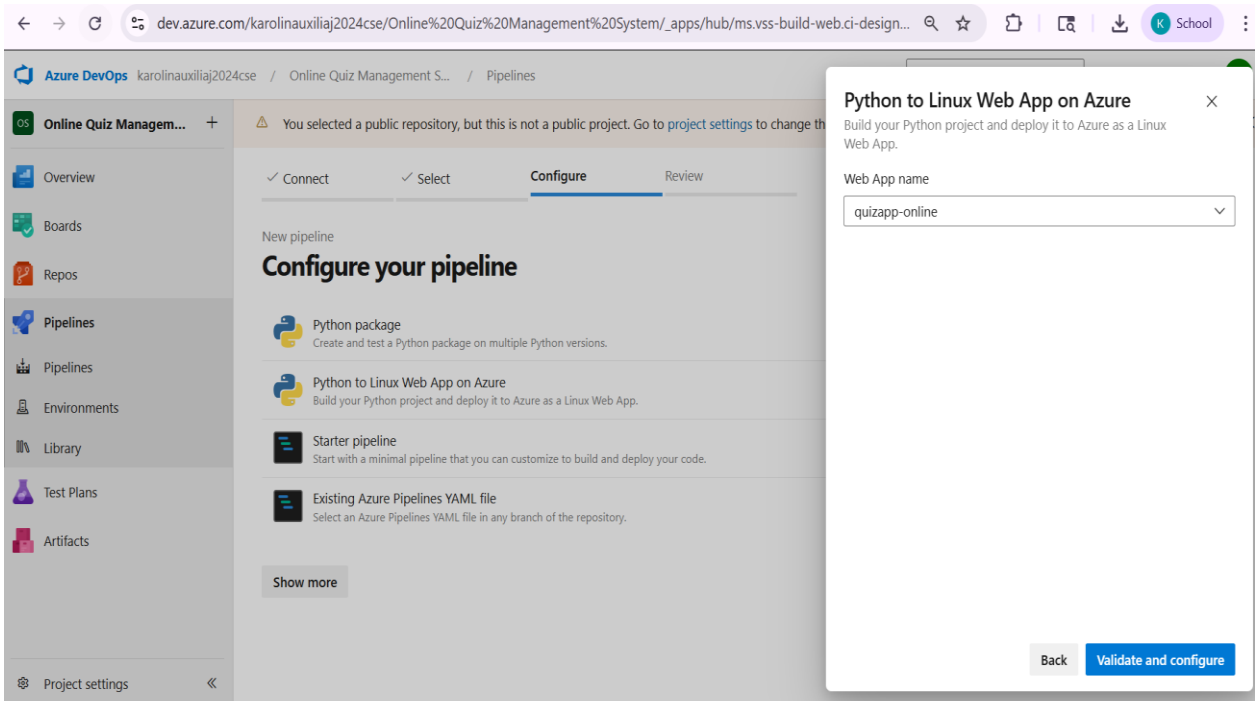
Continue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Show more
Python to Linux Web App on Azure
Build your Python project and deploy it to Azure as a Linux Web App.
Web App name
quizapp-online
Back
Validate and configure

dev.azure.com/karolinauxiliaj2024cse/Online%20Quiz%20Management%20System/_apps/hub/ms.vss-build-web.ci-design...

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Project settings

You selected a public repository, but this is not a public project. Go to project settings to change the visibility of the project. Learn more

Connect ✓ Select ✓ Configure Review

New pipeline

Review your pipeline YAML

Variables | Save and run

Karolin-Auxilia/Online-Quiz-Management-System / azure-pipelines.yml *

Show assistant

```yaml
# Python to Linux Web App on Azure
# Build your Python project and deploy it to Azure as a Linux Web App.
# Change python version to one thats appropriate for your application.
# https://docs.microsoft.com/azure/devops/pipelines/languages/python

trigger:
- main

variables:
  # Azure Resource Manager connection created during pipeline creation
  azureServiceConnectionId: '2d842cd1-a04c-4023-b321-487998b137fe'

  # Web app name
  webAppName: 'quizapp-online'

  # Agent VM image name
  vmImageName: 'ubuntu-latest'
```

dev.azure.com/karolinauxiliaj2024cse/Online%20Quiz%20Management%20System/_apps/hub/ms.vss-build-web.ci-design...

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New pipeline

Review your pipeline YAML

Karolin-Auxilia/Online-Quiz-Management-System / azure-pipelines.yml *

```yaml
    runOnce:
      deploy:
        steps:

          Settings
        - task: UsePythonVersion@0
          inputs:
            versionSpec: '$(pythonVersion)'
          displayName: 'Use Python version'

          Settings
        - task: AzureWebApp@1
          displayName: 'Deploy Azure Web App : quizapp-online'
          inputs:
            azureSubscription: $(azureServiceConnectionId)
            appName: $(webAppName)
            package: $(Pipeline.Workspace)/drop/$(Build.BuildId).zip
```

Save

Saving will commit azure-pipelines.yml to the repository.

Commit message

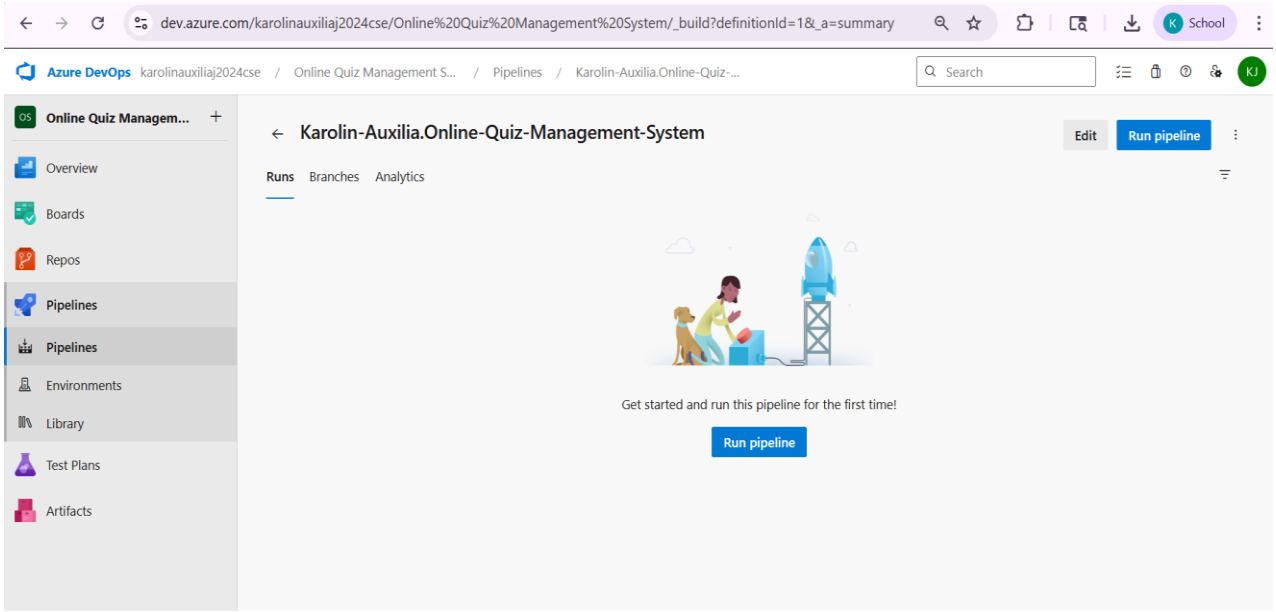
Set up CI with Azure Pipelines

Optional extended description

Add an optional description...

- (•) Commit directly to the main branch
- ( ) Create a new branch for this commit

Save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Edit
Run pipeline
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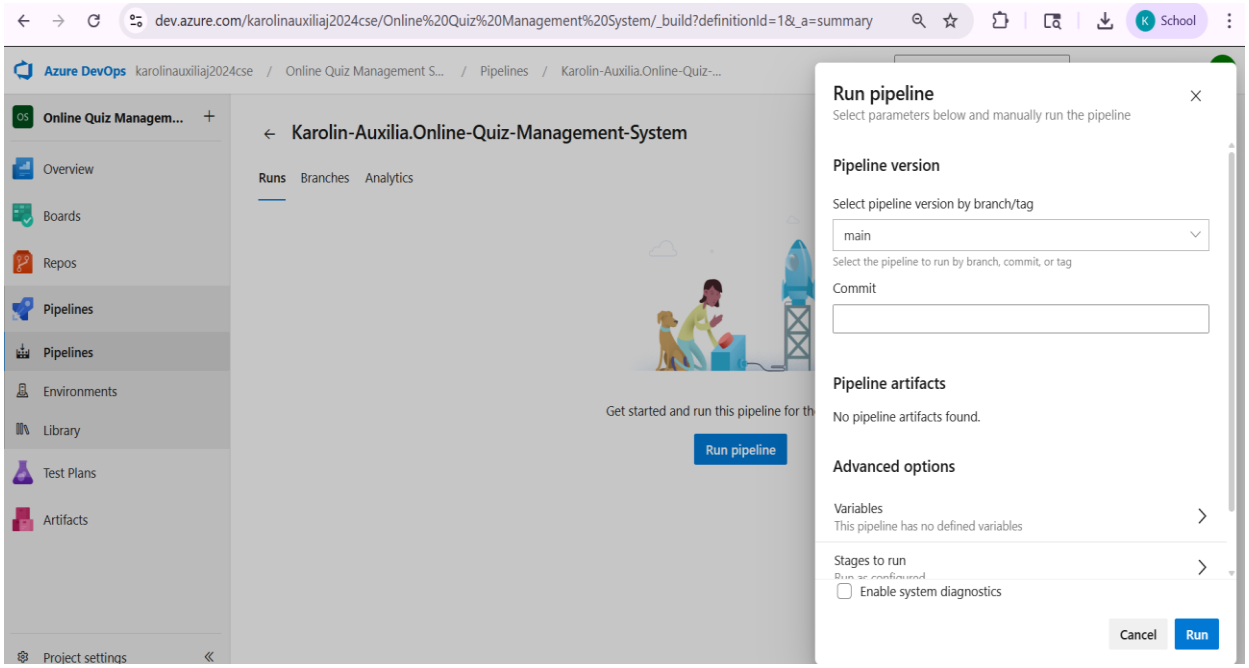
Analytics
Get started and run this pipeline for the first time!
Run pipeline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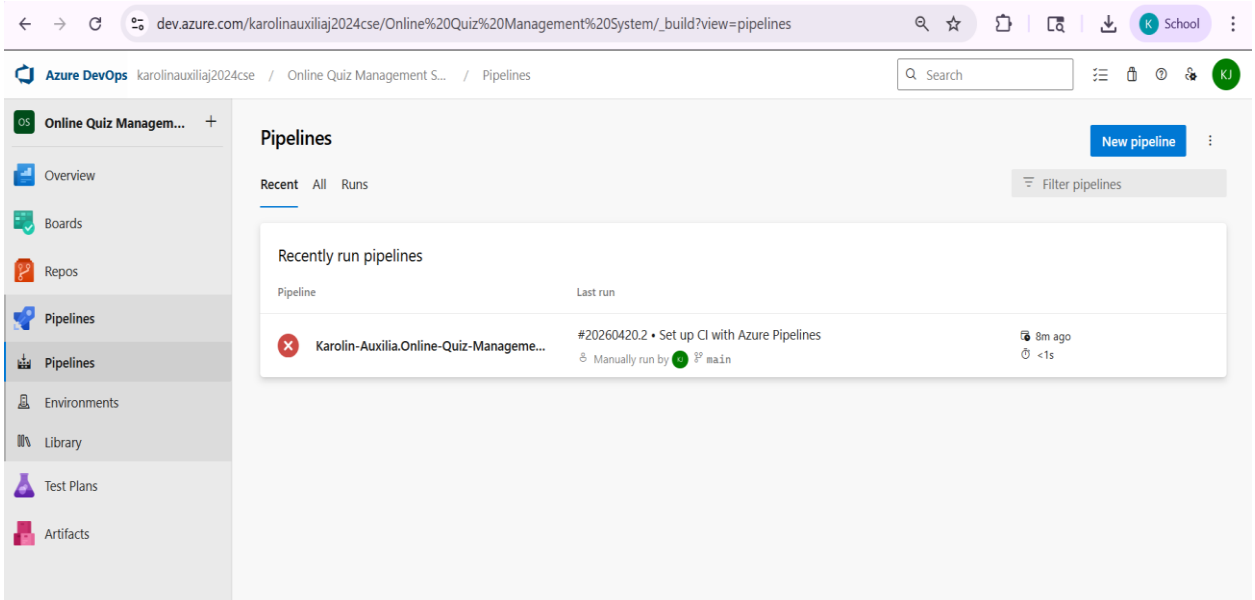
Analytics
Run pipeline
Run pipeline
Select parameters below and manually run the pipeline
Pipeline version
Select pipeline version by branch/tag
main
Select the pipeline to run by branch, commit, or tag
Commit
Pipeline artifacts
No pipeline artifacts found.
Advanced options
Variables
This pipeline has no defined variables
Stages to run
Enable system diagnostics
Cancel
Run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